

Large closed sets avoiding a continuum of power asymptotics

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Abstract

For every prescribed nonempty countable family of positive configurations with bounded gaps in their occupied dyadic logarithmic bins, and every $0 < \varepsilon < 1$, we construct a closed nowhere dense one-periodic set whose measure in every unit interval is strictly greater than $1 - \varepsilon$. It simultaneously misses infinitely many distinct values in every tail of every map with a nonzero leading power and a power-controlled higher order remainder. The positive leading exponent, translation, coefficient and remainder map all vary after the avoiding set has been chosen. In particular a compact subset of $[0, 1]$ of measure greater than $1 - \varepsilon$ avoids every affine null geometric progression, with every ratio in $(0, 1)$, in every tail. The proof uses finite random routing on nested periodic grids, exact representatives of a continuum of parameters including boundary strata, and open repair of the closed set of missed centers. A logarithmic window schedule makes the finer grid compatible with positive power remainders. The traditional proof is given in full; a separate correspondence records its relation to frozen Lean proofs.

1 Statement, quantifiers and context

Let μ be Lebesgue measure. All real powers below have positive bases. A set $S \subset \mathbb{R}$ is one-periodic if $S + 1 = S$; for a measurable such set write $\text{dens } S = \mu(S \cap [0, 1))$. The integral of a one-periodic indicator over any interval of length one is the same: after reducing its left endpoint modulo one, split the interval there and translate its second part by -1 . Endpoints have measure zero, so closed unit intervals give the same value.

Definition 1.1. For $A \subset (0, \infty)$ put

$$S(A) = \{j \geq 1 : A \cap (2^{-j-1}, 2^{-j}] \neq \emptyset\}.$$

We say that A is logarithmically syndetic near zero if there are integers $G, J \geq 1$ such that every $\{j, \dots, j + G - 1\}$ with $j \geq J$ meets $S(A)$.

The constants G, J may depend on the configuration. The sets A need not be measurable or countable. They have arbitrarily small positive members.

Theorem 1.2 (Prescribed family and continuum of power asymptotics). *Fix a nonempty countable index set Λ and configurations $(A_\ell)_{\ell \in \Lambda}$ that are logarithmically syndetic near zero. For every $0 < \varepsilon < 1$ there is a closed nowhere dense one-periodic $E \subset \mathbb{R}$ such that*

$$\mu(E \cap [x, x + 1]) > 1 - \varepsilon \quad (x \in \mathbb{R}). \quad (1)$$

The same E has the following property for every $\ell \in \Lambda$, $s > 0$, $\alpha > 0$, $y \in \mathbb{R}$, $c \neq 0$, and finite $M \geq 0$. Let $\delta > 0$ and let f be defined on $D = A_\ell \cap (0, \delta)$. Suppose that, for some $\rho_0 > 0$,

$$|f(a) - y - ca^s| \leq Ma^{s+\alpha} \quad (a \in D, 0 < a < \rho_0). \quad (2)$$

Then for every $\rho > 0$ the set

$$\{f(a) : a \in D, a < \rho, f(a) \notin E\} \quad (3)$$

is infinite. No continuity, measurability or injectivity of f is assumed.

Thus the family precedes the choice of E , while all the parameters and maps in (2) follow it. The theorem does not choose one set for all possible configurations. Infinitude refers to distinct output values, rather than merely infinitely many input indices. The eventual hypothesis permits partial functions on a tail without any extension regularity requirement.

Corollary 1.3 (Compact version). *Under the hypotheses of Theorem 1.2 there is a compact nowhere dense $F \subset [0, 1]$ with $\mu(F) > 1 - \varepsilon$ for which (3), with E replaced by F , is infinite for all the stated maps and parameters.*

Corollary 1.4 (All ratios of affine null geometric progressions). *For every $0 < \varepsilon < 1$ there is a compact nowhere dense $F \subset [0, 1]$ with $\mu(F) > 1 - \varepsilon$ such that*

$$\forall a \neq 0 \forall b \in \mathbb{R} \forall q \in (0, 1) \forall N \in \mathbb{N}, \quad \{aq^n + b : n \geq N\} \setminus F \text{ is infinite.}$$

In particular no tail of any such progression is contained in F .

The proof of the main theorem occupies Sections 2–8. The corollaries are proved in Section 8.

1.1 Related work and the role of the normalization

Burgin, Goldberg, Keleti, MacMahon and Wang [1] ask in their Question 1 whether every positive measure measurable subset of $\mathbb{R}$ contains some $(aq^n + b)_{n \geq 1}$, with $a \neq 0$ and $0 < q < 1$. Their Theorem 1.1 constructs a compact set with zero as a density point avoiding all nonconstant *untranslated* $(aq^n)_{n \geq 1}$. The translation in their question is therefore material. Corollary 1.4 supplies a negative answer to that question, with the stronger measure and tail conclusions stated above. This is a logical comparison of the exact statements, not a claim that a literature search certifies exclusive priority.

Kolountzakis and Papageorgiou [2], Theorem 4.1, use finitely many representatives of transformation parameters obtained from grid-crossing hyperplanes. Their work provides precedent for probabilistic grid avoidance and polynomial parameter arrangements; its main setting concerns sequences escaping to infinity under growth hypotheses. The planar sign-arrangement argument here is elementary and is proved below. The power normalization $q = 2^{-s}$ and the dyadic coefficient normalization $|c| = 2^k t$ are also elementary reductions, not separate contributions.

The construction integrates the nested-grid routing and exceptional-center repair recorded in the pinned geometric-case research source [4] and in the nonlinear avoidance entry [5]. The prescribed countable logarithmic profile perspective of the latter motivates working in output logarithmic windows. The former chooses its avoiding set for a fixed ratio; the latter prescribes a countable profile family and permits positive logarithmic upper Banach density. We impose the stronger bounded-gap configuration hypothesis while allowing a continuum of leading exponents; neither parameter scope nor configuration scope should be conflated. The specific extension proved here handles a continuum of exponents *inside* each finite routing lemma: active input indices depend on the exponent, exact affine signs determine their grid addresses, and a coupled logarithmic-window and power-buffer schedule keeps the error cost small. We give the conditioning and repair arguments anew, so no theorem from those research sources is a premise of the traditional proof.

The power gain $\alpha > 0$ is a substantive boundary. Feng, Lai and Xiong [3], Theorem 1.1, show that positive decreasing sequences with $\limsup a_{n+N}/a_n < 1$ for some N embed into every positive measure set by a bi-Lipschitz map differentiable at zero with derivative one. For a dyadic sequence this gives a remainder $o(a)$, without a promised $O(a^{1+\alpha})$ rate. Accordingly Theorem 1.2 asserts avoidance of its specified power-controlled maps, not arbitrary differentiable, C^1 , or flat germs. The source-specific comparisons and their limitations are documented in the supplement; no exhaustive novelty or current-status certification is asserted.

2 The robust compact-parameter blocker

Lemma 2.1 (Uniform robust blocker). *Fix a logarithmically syndetic $A \subset (0, \infty)$, real numbers $0 < s_0 < s_1$, $\alpha_0 > 0$, integers $k \in \mathbb{Z}$, $q, h \geq 1$, and $0 < p < 1/12$. There is an open one-periodic $H \subset \mathbb{R}$ with $\text{dens } H \leq 6p$ such that for every $x \in \mathbb{R}$, $s \in [s_0, s_1]$, $t \in [1, 2]$, and arbitrary errors $e : A \cap (0, 2^{-h}) \rightarrow \mathbb{R}$ satisfying*

$$|e(a)| \leq qa^{s+\alpha_0},$$

some $a \in A \cap (0, 2^{-h})$ satisfies

$$x + t2^k a^s + e(a) \in H. \quad (4)$$

The set H is chosen before x, s, t, e .

The next six sections prove this lemma. All probability spaces used are finite. The errors in its statement are deterministic arbitrary choices; no independence assumption about them is involved.

Lemma 2.2 (Separated sampling and open-window counts). *There are $a_n \in A \cap (0, 2^{-h})$ decreasing to zero, and $B_0 > 0$, such that with $z_n = -\log_2 a_n$,*

$$3/s_0 < z_{n+1} - z_n < B_0. \quad (5)$$

Put $D = s_1 B_0$ and $\eta = 1/(2D)$. For every $s \in [s_0, s_1]$ the sequence $w_n(s) = sz_n - k$ increases to infinity with successive gaps greater than 3 and less than D . If $u \geq s_1 z_1 - k$ and $L \geq 2D$, then

$$\eta L \leq \#\{n : u < w_n(s) < u + L\} \leq 1 + L/3. \quad (6)$$

The inequalities apply at parameters where a member equals a window endpoint.

Proof. Choose an occupied bin $j_1 > \max(J, 4, h)$, and let $m = \lceil 3/s_0 \rceil + 2$. Having chosen j_n , syndeticity permits an occupied $j_{n+1} \in \{j_n + m, \dots, j_n + m + G - 1\}$. Choose a_n in its bin. Then $z_n \in [j_n, j_n + 1)$, so $z_{n+1} - z_n > m - 1 > 3/s_0$ and $z_{n+1} - z_n < m + G$. Take $B_0 = m + G$. These bounds also prove convergence and the assertions about $w_n(s)$. There is a last member at or below u and a first member at or above $u + L$. If r members lie strictly between these endpoints, the length L is covered by $r + 1$ successive gaps, giving $L \leq (r + 1)D$. Thus $r \geq L/D - 1 \geq \eta L$. Separation greater than 3 gives the upper bound. Equality with either endpoint is deliberately excluded from r . $\square$

3 Routing windows and actual periodic grids

Fix the data of Lemma 2.1 and the sample above. Choose a branching integer $b \geq 2$ such that $p\eta(b - 1) \geq 12$, and put

$$\lambda = p\eta(b - 1)/2, \quad \kappa = \lambda - 4 \log 2 > 0. \quad (7)$$

Here and subsequently unmarked logarithms are natural. Choose $d \geq 1$ so that $(1 - 2^{1-b})^d < p$. Let K be the number of edges in the complete ordered b -ary tree of depth d . Choose an integer $g \geq 1$ with

$$K2^{3-g} < p. \quad (8)$$

All of $b, d, K, g, \eta, \lambda, \kappa$ are fixed before choosing the late position of the routing template.

Fix integers U, L satisfying the standing conditions

$$U \geq 4, \quad U > |k|, \quad U \geq s_1 z_1 - k, \quad L \geq \max\{2, \lceil 2D \rceil\}.$$

These conditions apply throughout the routing and probability arguments below; Section 6 will impose a sufficiently late schedule on the same parameters. For this base length L , define

$$\begin{aligned} L_1 &= L, & T_1 &= bL + (b-1)g, \\ L_r &= g + T_{r-1}, & T_r &= b(L_r + g + T_{r-1}) + (b-1)g \quad (r \geq 2). \end{aligned} \quad (9)$$

The height of a vertex is its remaining distance to the leaves. Each of its outgoing edges receives an integer window of length L_r when its height is r . Traverse edges in preorder, including default edges, placing g unused integer indices between consecutive edge windows. Start at this integer U . For edge e write its endpoints $u_e, v_e = u_e + L_r - 1$, and write v_e^* for the last endpoint of that edge and its entire child subtree. Set $T = T_d$.

Lemma 3.1 (Template spans). *The last endpoint of the template is $U + T - 1$, every edge length is at least L , and for an edge of length L_r ,*

$$v_e^* - u_e + 1 \leq 2L_r. \quad (10)$$

For fixed b, d, g , the total span T is affine in L with positive finite integer coefficients.

Proof. A height-one subtree has the b windows of length L separated by $b-1$ gaps. At height $r \geq 2$ one edge followed by its child subtree has span $L_r + g + T_{r-1} = 2L_r$. There are b such pieces and $b-1$ intervening gaps, giving (9). This also proves the endpoint formula by induction. At height one the local span is only L_1 . Finally $T_r = 2bT_{r-1} + (3b-1)g$ for $r \geq 2$, proving the affine assertion. $\square$

For every integer endpoint v use the periodic grid of $N_v = 2^{v+3}$ cells, with address

$$J_v(x) = \lfloor N_v x \rfloor \pmod{N_v}.$$

Its cells are left closed and right open. The grids are nested. At each nondefault edge (child number $i < b$) put a table of independent fair selector bits indexed by its endpoint-grid cells. At each leaf put an independent Bernoulli- p terminal table indexed by the endpoint grid of its incoming edge. All entries of all tables are mutually independent. Route a point from the root by taking the first nondefault child with selector bit one at its address, or child b if all the nondefault bits are zero. Include the point in B when its terminal bit is one. For each table outcome B is a measurable periodic union of cells of the single finest grid

$$N = 2^{U+T+2}. \quad (11)$$

For a fixed point, conditioning on every selector leaves its addressed terminal entry Bernoulli- p . Finite summation followed by integration over $[0, 1)$ gives

$$\mathbb{E} \text{dens } B = p. \quad (12)$$

An index n is *active* on edge e at exponent s precisely when

$$u_e < w_n(s) < v_e + 1 = u_e + L_r. \quad (13)$$

These are fixed *output* windows. The input indices supplying their tests may vary with s . This feature avoids requiring one input window to separate outputs at s_1 while preserving their earlier addresses at s_0 .

Call a center x stable if, for every noninitial window starting at $u_e = v + g + 1$, no boundary of the grid N_v lies in $(x, x + 2^{-v-g}]$. This is a deterministic measurable periodic condition.

Lemma 3.2 (Stability and address separation). *The unstable centers have density at most $K2^{3-g} < p$. Assume $U \geq 4$. For a stable x , any active point $x + t2^k a_n^s$, $t \in [1, 2]$, has all earlier preorder-grid addresses equal to those of x . On its own edge grid its address differs from the center's, and two distinct active indices on that edge have different addresses. These distinctions persist on every finer grid.*

Proof. For a preceding endpoint v , the bad-center intervals have total length at most $N_v 2^{-v-g} = 2^{3-g}$ per period. Sum over at most K predecessors. An active offset satisfies

$$2^{-u_e - L_r} < t2^{k - sz_n} < 2^{1 - u_e}. \quad (14)$$

If there is a predecessor v , the upper bound is 2^{-v-g} , so stability prevents crossing its boundary. All earlier grids divide that grid; hence their addresses agree as well. The own grid has width $2^{-u_e - L_r - 2}$, equal to one quarter of the displayed lower offset bound. Different active logarithms have gap greater than 3, so their positive offsets have ratio greater than 8. Their difference is greater than $72^{-u_e - L_r}$, also exceeding the cell width. All offsets lie in an interval of length less than $1/8$ since $U \geq 4$, so the separations cannot be hidden by periodic wraparound. Nested finer cells preserve the distinctions. Floor addresses at exact grid boundaries obey the same left-closed convention throughout. $\square$

4 Finite probability with center exposure

Fix a stable center x . Expose its addressed selector entry in *every* selector table, not only the tables on its route. These entries determine its route and partition the finite probability space into center-exposure atoms. The probability that this route never takes a default edge is

$$(1 - 2^{1-b})^d < p. \quad (15)$$

Indeed at each reached vertex the fresh $b - 1$ exposed fair entries are all zero with probability 2^{1-b} , independently of the preceding route choices.

Consider an exposure atom whose first default is at a height- r vertex. Fix $(s, t) \in [s_0, s_1] \times [1, 2]$. For each outgoing nondefault edge e_i and each of its active indices, take a local test: its own selector bit times the terminal bit obtained by routing the point within child subtree i . By Lemma 3.2 the point agrees with the center at strict ancestors and at preceding siblings of this default vertex. They have the requisite exposed zero bits at those siblings. If its own selector is one and its local terminal bit is one, it belongs to the actual global set B .

Lemma 4.1 (Fixed-parameter joint miss). *Under the standing conditions of Section 3, on each first-default exposure atom, at fixed (s, t) , the probability that all these local tests fail is at most $\exp(-\lambda L_r)$. Equivalently, without dividing by the mass of the atom, its intersection with the miss event has probability at most its own mass times $\exp(-\lambda L_r)$.*

Proof. The number m_* of tests is at least $\eta(b - 1)L_r$ by Lemma 2.2, since $u_e \geq U \geq s_1 z_1 - k$ and $L_r \geq L \geq \lceil 2D \rceil$ on every outgoing edge. Their own selector coordinates are distinct and avoid every exposed coordinate: for one edge this follows from its own address separation, and different edges have different tables. Thus the tested bits $A_1, \dots, A_{m_*}$ are free independent fair bits after center exposure.

Condition next on *all* selector entries. Local terminal addresses are pairwise distinct for every such assignment. Points in different child subtrees or different leaves use different terminal tables. Points ending at the same leaf started on the same child edge, where their own-grid addresses were distinct, and keep distinct addresses at that leaf's finer grid. Hence terminal independence gives conditional failure probability $(1 - p)^{\sum_j A_j}$. Averaging over the free own-selector bits yields exactly $(1 - p/2)^{m^*}$, irrespective of shared auxiliary selector reads or changes in the terminal addresses. It is at most $\exp(-pm_*/2) \leq \exp(-\lambda L_r)$. Multiplying by exposure-atom masses and using finite sums proves the joint form, including zero-mass atoms. No independence of entire routing paths is assumed. $\square$

5 Exact representatives of the continuum

We give a boundary-complete finite reduction at a fixed center and fixed first-default vertex. It depends on the prescribed sample and actual grids, and is chosen before the unexposed tables.

Lemma 5.1 (Planar sign representatives). *For m affine real functions on $\mathbb{R}^2$, at most $2m^2 + 1$ different vectors of signs in $\{-1, 0, 1\}^m$ are realized. Consequently any subset of the plane admits a choice of at most $20(m + 5)^2$ representatives realizing all its sign vectors. All zero signs are retained.*

Proof. Constant functions have fixed signs. Merge coincident zero lines of nonconstant functions, keeping their individual oriented labels. With $r \leq m$ distinct lines there are at most $r(r - 1)/2$ intersection vertices. Each line is cut by at most $r - 1$ distinct intersection points into at most r open edges, giving at most r^2 edges. Insert the lines one by one; the j th line is cut into at most j pieces, each of which increases the number of open faces by at most one. There are therefore at most $1 + r(r + 1)/2$ faces. These strata cover the plane and have constant full sign vectors. Their total is at most $2r^2 + 1$. Intersecting with an arbitrary prescribed subset introduces no new sign vectors: choose one point for each vector realized in that subset. Parallel lines, repeated intersections and constant zero functions only reduce these counts. The larger displayed representative bound is convenient below. $\square$

Let $L_r = \ell$ at the first-default vertex, and suppose $U > |k|$. An index potentially active on edge e must have

$$(u_e + k)/s_1 \leq z_n \leq (u_e + \ell + k)/s_0.$$

Since all $z_n > 0$ have gaps greater than $3/s_0$, the number of candidates on an edge is at most $\lceil (U + T + |k|)/3 \rceil + 1$. Thus the number of candidate edge-index pairs is bounded by

$$P = (b - 1)(\lceil (U + T + |k|)/3 \rceil + 1). \quad (16)$$

For each pair record two activation cuts $sz_n - k = u_e, u_e + \ell$. Record also every lifted boundary of its *local finest* grid, namely the grid at v_e^* , whose offset from x is $\beta = j/N_{v_e^*} - x \in (0, 2^{1-u_e})$. By (10) there are at most

$$1 + N_{v_e^*} 2^{1-u_e} \leq 1 + 2^{2\ell+3}$$

such boundaries. Boundary crossing is the equality $t2^{k-sz_n} = \beta$. In coordinates $(s, v) = (s, \log t)$ it is the affine cut

$$v - sz_n \log 2 + k \log 2 - \log \beta = 0. \quad (17)$$

The sign equals the sign of the output's difference from that lifted grid boundary, because logarithm is strictly increasing. The activation signs and the signs in (17) therefore determine the vector that gives *none* for inactive pairs and the actual local finest-grid address for active ones. That grid determines every table read in the pair's subtree.

There are at most $m = P(3 + 2^{2\ell+3})$ cuts. Apply Lemma 5.1 to the closed rectangle $[s_0, s_1] \times [0, \log 2]$ and pull representatives back to (s, t) . The resulting set has size at most

$$20[P(3 + 2^{2\ell+3}) + 5]^2. \quad (18)$$

At an activation equality that pair is inactive. At a grid equality it has the floor key of the cell to the right. The sign zero records both conventions exactly. Parameter endpoints, intersections, coincident cuts and lower dimensional strata are covered by the same argument. In particular this is not a numerical sampling of exponents, and the representatives work for *every* assignment of all remaining table entries.

Lemma 5.2 (Continuum miss bound). *Under the same standing conditions, let C be a first-default center-exposure atom, and let $\mathcal{M}_C$ be the event that, at its first-default vertex, all active local tests fail for some $(s, t) \in [s_0, s_1] \times [1, 2]$. Then the joint probability satisfies*

$$\mathbb{P}(C \cap \mathcal{M}_C) \leq \mathbb{P}(C) 5120b^2(U + T + |k| + 2)^2 e^{-\kappa\ell}. \quad (19)$$

Proof. Every miss has the same active-address vector as a retained representative, and therefore the same local tests for every outcome. Use the finite union bound, (18), and Lemma 4.1 at each representative. The factor multiplying $\mathbb{P}(C)$ is $20[P(3 + 2^{2\ell+3}) + 5]^2 e^{-\lambda\ell}$. For $P \geq 1$ and $\ell \geq 2$,

$$P \leq b(U + T + |k| + 2), \quad P(3 + 2^{2\ell+3}) + 5 \leq 16P2^{2\ell}.$$

Squaring and using (7) gives (19). The factor depending on the absolute position U has thus been retained. Summing the joint estimates over atoms avoids division by any atom mass. $\square$

6 A coupled window and error-buffer schedule

For integer U define

$$L(U) = \max \left\{ g, \lceil 2D \rceil, 2, \left\lceil \frac{4 \log(U + 2) + \log(46080b^2/p) + 1}{\kappa} \right\rceil \right\}. \quad (20)$$

Use this length in (9), writing the resulting span as $T(U)$. By Lemma 3.1, $T(U) = O(\log U)$, with a finite constant depending on the already fixed tree. In particular $T(U)/U \rightarrow 0$.

Lemma 6.1 (Simultaneous late schedule). *There are arbitrarily large integers U for which*

$$U \geq 4, \quad U > |k|, \quad U \geq s_1 z_1 - k, \quad T(U) \leq U, \quad (21)$$

every first-default continuum miss bound is less than p times its atom mass, and a common positive error radius r_U has $4Nr_U < p$, where N is the actual finest grid (11).

Proof. The conditions (21) hold for all sufficiently large U . For $\ell \geq L(U)$, $T(U) \leq U$, and $|k| < U$, the factor multiplying the atom mass in (19) is at most

$$46080b^2(U + 1)^2 e^{-\kappa L(U)} \leq p e^{-1} \frac{(U + 1)^2}{(U + 2)^4} < p. \quad (22)$$

For any active input on any edge, $sz_n - k > U$ and $s \leq s_1$ give $2^k a_n^s < 2^{-U}$ and $z_n > (U + k)/s_1$. Since $U + k > 0$, its allowed error has the bound

$$q a_n^{s+\alpha_0} \leq q 2^{-k} 2^{-U} 2^{-\alpha_0(U+k)/s_1}.$$

Define

$$r_U = (q2^{-k} + 1)2^{-U}2^{-\alpha_0(U+k)/s_1} > 0. \quad (23)$$

The allowed active error is strictly smaller than r_U . The doubled-buffer cost on the *actual* grid is

$$4Nr_U = 2^{T(U)+4}(q2^{-k} + 1)2^{-\alpha_0(U+k)/s_1} \longrightarrow 0. \quad (24)$$

Indeed its logarithm is $O(\log U) - \alpha_0(\log 2)U/s_1 + O(1)$, which tends to $-\infty$. Increasing U further enforces $4Nr_U < p$ while keeping all preceding constraints. The order is noncircular: tree constants are fixed, then $L(U), T(U)$ are explicit logarithmic functions, and all inequalities hold eventually for those same functions. No practical bound on the size of the finite probability space is asserted. $\square$

7 Closed missed centers and open repair

Fix a schedule from Lemma 6.1. For each outcome put

$$B_1 = B + (-r_U, r_U), \quad B_2 = B + (-2r_U, 2r_U).$$

These are open periodic sets. A union of cells of an N -cell periodic grid enlarged by radius r gains density at most $2Nr$: enlarge each of its at most N cell intervals by r at both ends and use subadditivity, allowing periodic overlap. Thus (12) and (24) give

$$\mathbb{E} \text{dens } B_2 \leq p + 4Nr_U < 2p. \quad (25)$$

Only finitely many indices can be active on some template edge for some $s \in [s_0, s_1]$, since their logarithms lie in $[(U+k)/s_1, (U+T+k)/s_0]$ and are separated. Let I contain precisely these potential indices. For $n \in I$ let S_n be the relatively open subset of $[s_0, s_1]$ where $w_n(s)$ belongs to the union of the open template windows. Define R to be the set of centers for which some (s, t) in the compact parameter rectangle has

$$s \notin S_n \quad \text{or} \quad x + t2^k a_n^s \notin B_1 \quad \text{for every } n \in I. \quad (26)$$

Lemma 7.1 (Closedness and expected density of missed centers). *For each outcome R is closed and one-periodic, and $\mathbb{E} \text{dens } R \leq 3p$.*

Proof. For each n , the first alternative in (26) is closed because S_n is relatively open. The second is closed by continuity of $(x, s, t) \mapsto x + t2^k a_n^s$ and openness of B_1 . Their union and the finite intersection over I are closed. Projection along a compact parameter space preserves closedness: from a convergent sequence of projected centers choose parameter witnesses, pass to a convergent subsequence in the rectangle, and use closedness. Translation of x by an integer preserves all conditions. Thus R is measurable as well.

At a stable center, membership in R implies a continuum local miss at its first-default vertex, unless the center route has no default. This follows because a successful local test gives membership in $B \subset B_1$. Sum (22) over the first-default atoms and add (15); the pointwise probability is at most $2p$. At an unstable center it is at most one. Finite summation and integration over a period, together with (8), give

$$\mathbb{E} \text{dens } R \leq 2p\mu(\text{stable} \cap [0, 1)) + \mu(\text{unstable} \cap [0, 1)) \leq 3p.$$

All exchanges are finite sums of measurable indicators. No measurable choice of continuum representatives as a function of x is needed for this pointwise probability estimate. $\square$

Lemma 7.2 (Deterministic repair). *There is a table outcome and an open periodic $V \supset R$ for which $H = B_2 \cup V$ has $\text{dens } H \leq 6p$ and satisfies (4) for every center and every permitted error family.*

Proof. By (25) and Lemma 7.1, some outcome has $\text{dens } B_2 + \text{dens } R \leq 5p$. To justify the existence directly, a finite weighted average with nonnegative weights summing to one cannot be smaller than every positive-weight value; zero-weight outcomes cause no problem.

For this closed periodic R , the open thickenings $V_j = \{x : \text{dist}(x, R) < 1/j\}$ decrease to R , are periodic, and their measures restricted to $[0, 1)$ decrease to $\text{dens } R$ by continuity from above in a finite measure space. (If R is empty use $V = \emptyset$.) Choose j with $\text{dens } V_j < \text{dens } R + p$ and set $V = V_j$. Consequently $\text{dens}(B_2 \cup V) < 6p$.

Fix x, s, t, e . If $x \notin R$, the negation of (26) at these parameters supplies a finite active input with unperturbed point in B_1 . Its error has absolute value at most r_U , so its perturbed point belongs to B_2 : a distance strictly less than r_U from B remains strictly less than $2r_U$ after an error of size at most r_U . Equality of an allowed error does not lose an open-buffer endpoint.

If $x \in R \subset V$, use the full sampled sequence. Its perturbed offsets tend to zero; for $a_n < 1$,

$$|t2^k a_n^s + e(a_n)| \leq 2^{k+1} a_n^{s_0} + q a_n^{s_0 + \alpha_0} \rightarrow 0.$$

Since V is open and contains x , a sufficiently late perturbed point lies in V . This index need not lie in I or be active on a finite window. Every sampled input belongs to $A \cap (0, 2^{-h})$. Thus every center is repaired, including all exceptions, and Lemma 2.1 follows. $\square$

8 Countable exhaustion and distinct tail values

Proof of Theorem 1.2. The set of tuples

$$\tau = (\ell, N, j, k, q, h), \quad \ell \in \Lambda, \quad N \geq 2, \quad j, q, h \geq 1, \quad k \in \mathbb{Z},$$

is countable. Choose positive $p_\tau < 1/12$ such that $12 \sum_\tau p_\tau < \varepsilon$; for example, using an injection of the tuples into $\mathbb{N}$, assign $p_\tau = \varepsilon 2^{-n(\tau)-1}/24$. For each tuple apply Lemma 2.1 to A_ℓ with $[s_0, s_1] = [1/N, N]$ and $\alpha_0 = 1/j$. Let H_τ be its blocker and put

$$\mathcal{U} = \bigcup_\tau (H_\tau \cup (-H_\tau)), \quad E = \mathbb{R} \setminus \mathcal{U}.$$

This is a countable union of open periodic sets with $\text{dens } \mathcal{U} \leq 12 \sum_\tau p_\tau < \varepsilon$. Thus E is closed and periodic, and the unit-interval identity in Section 1 gives the strict inequality (1).

Now fix the configuration, map, and parameters in (2). Choose N with $s \in [1/N, N]$, j with $1/j \leq \alpha$, and an integer $q \geq \max(1, M)$. For $a < 1$ the remainder bound is at most $q a^{s+1/j}$. If $c > 0$ choose $k \in \mathbb{Z}$ and $t \in [1, 2)$ with $c = 2^k t$. For every sufficiently large h , $2^{-h} < \min(1, \delta, \rho_0)$. Apply the corresponding blocker to $x = y$ and the errors $e(a) = f(a) - y - c a^s$ on this tail. It gives $a_h \in A_\ell$, $a_h < 2^{-h}$, with $f(a_h) \in H_\tau \subset \mathcal{U}$. If $c < 0$, apply the same argument to $-f$, centered at $-y$, with coefficient $-c > 0$ and errors $-e(a)$; then $f(a_h) \in -H_\tau \subset \mathcal{U}$. This uses only countably many budgets even though s, α, c, y, f range after the set is chosen.

Since $M a^\alpha \rightarrow 0$ and $c \neq 0$, sufficiently small inputs satisfy

$$\frac{1}{2} |c| a^s \leq |f(a) - y| \leq \frac{3}{2} |c| a^s. \quad (27)$$

For any given $\rho > 0$, choose hits above with h arbitrarily large and $a_h < \rho$. Their values lie outside E , differ from y , and tend to y . They cannot form a finite set: a finite set avoiding y has a positive minimum distance from y , contrary to the upper bound in (27). This proves

infinitude of distinct values in every tail, without assuming distinctness of chosen indices or injectivity of f .

Finally E has empty interior. If it contained an open interval, choose a point y strictly inside that interval and any configuration in the nonempty family. The map $f(a) = y + a$ on a sufficiently small tail would take all its values inside the interval. It has $s = 1, c = 1, M = 0$ and any $\alpha > 0$, in contradiction with (3). A closed set with empty interior is nowhere dense. This completes the theorem. $\square$

Proof of Corollary 1.3. Take $F = E \cap [0, 1]$. It is compact, has measure strictly greater than $1 - \varepsilon$, and has empty interior because it is a subset of E . Every value outside E is outside F , so each infinite tail conclusion passes to F . $\square$

Proof of Corollary 1.4. Use the singleton family $A = \{2^{-n} : n \geq 1\}$. Every bin in Definition 1.1 contains its upper endpoint, so $G = J = 1$ works. For $q \in (0, 1)$ set $s = -\log_2 q > 0$, $c = a$, and $y = b$. The map $f(u) = b + au^s$ has zero remainder and gives $f(2^{-n}) = b + aq^n$. A sufficiently small input tail forces $n \geq N$. Corollary 1.3 gives infinitely many distinct misses, simultaneously for all the indicated parameters. In the separately frozen geometric proof, the endpoint asks only for at least one miss in every index tail; the present statement implies that endpoint and contains it as a corollary, without counting it as another independent contribution. $\square$

9 Boundaries and an obstruction to this mechanism

The theorem requires $s > 0$, $\alpha > 0$, and $c \neq 0$. A zero leading coefficient does not force nonconstant values, and positive α is what makes (24) tend to zero with a logarithmic total span. The proof does not extend by merely replacing this gain with an arbitrary $o(a^s)$ bound. The compact parameter rectangle includes its entire boundary; activations are open, while grid cells are left closed. These choices play different roles.

Here are two exact finite controls illustrating those roles. They supplement, but do not replace, the infinite proof. For inputs $a_1 = 1/2, a_2 = 1/4$, $s, t \in [1, 2]$, and holes $H = (1/10, 1) + \mathbb{Z}$, arbitrary errors of size at most $r < 7/160$ cannot keep both outputs outside H , whatever the center x . Indeed with $u = 2^{-s}$ their positive unperturbed difference is $t(u - u^2) \in [3/16, 1/2]$, and after errors the difference lies strictly between $1/10$ and $9/10$. Two points in the closed complementary gaps cannot have such a difference. At $r = 7/160$ take $s = 2, t = 1, x = 143/160$ and errors $(-7/160, 7/160)$: the perturbed points are $11/10$ and 1 , both missed endpoints. For $s < 2$ or $t > 1$ the same worst error bound still succeeds. Deleting the parameter boundary therefore loses this sharp failure.

Closed activation would also invalidate the closed-projection argument. For one input $a = 1/2$, $x = 0, t = 1$, ambient exponent interval $[1/2, 5/2]$, closed activation $s \in [1, 2]$ and holes $(2/5, 3/5)$, the condition “inactive or missed” holds along $s < 1$ but fails at the limit $s = 1$. With open activation $(1, 2)$ the limit is inactive and the relevant union is closed, as used in Lemma 7.1.

Proposition 9.1 (Positive upper Banach density does not suffice for the window-counting step). *There is a set $S \subset \mathbb{N}$ of upper Banach density one such that, for $A = \{2^{-j} : j \in S\}$ and each fixed finite $C > 0$, every sufficiently large U and every $0 < T \leq C \log U$ admit an $s \in [1, 2]$ for which*

$$\{j \in S : U < sj < U + T\} = \emptyset.$$

This is an obstruction to the present uniform output-window mechanism, not a disproof of an avoidance theorem for that larger class of configurations.

Proof. Put $R_m = 2^{2^m}$ and $S = \bigcup_{m \geq 3} \{R_m, R_m + 1, \dots, R_m + m\}$. Its blocks have arbitrary length, so for every integer interval length the maximum occupied proportion is one, giving upper Banach density one. An active j for $s \in [1, 2]$ must lie in $[U/2, U + T]$. For large U we have $T \leq U/4$. Any block meeting that interval has $R_m \leq 5U/4$, hence $m = O(\log \log U)$, and $R_m + m \geq U/2$, hence eventually $R_m \geq U/3$. There is at most one such block: consecutive starts obey $R_{m+1} = R_m^2$, so two starts in $[U/3, 5U/4]$ would require $U^2/9 \leq 5U/4$. If no block meets the interval, any s works. Otherwise the exponents that activate any member of that block lie in

$$\left(\frac{U}{R_m + m}, \frac{U + T}{R_m} \right).$$

The length is at most $3T/U + 9m/U$, tending to zero uniformly for the stated T . It is eventually less than one and cannot cover $[1, 2]$. Choose an exponent outside it. $\square$

A Formal correspondence and evidence status

The formal sources and their dependency locks are preserved as two immutable projects in the public repository: `formalizations/geometric-avoidance` and `formalizations/continuum-remainder-avoidance`. The stronger endpoint is `ContinuumRemainder.continuum_power_target`; its compact endpoint is `ContinuumRemainder.compact_power_avoidance`. The earlier endpoint is `ContinuumGeometric.geometric_main_target`. The formal definitions use total functions with an eventual bound. A function on $A \cap (0, \delta)$ can be extended arbitrarily outside that domain; the exact tail value sets below δ are unchanged. Conversely restriction of a total function gives the partial-domain statement. The corresponding Lean declarations are listed in the table. This is equivalence of the tail quantifiers, not a continuity assertion about an extension.

The 18 stronger proof modules reuse 45 geometric modules byte for byte. Two additional aggregate modules account for the independent audit’s reported 65-module count. The complete machine-readable declaration inventory includes source locations and hashes; the module import graph is provided separately. Those import edges record dependencies between modules, not an asserted exact constant-level kernel dependency graph. Every key proved step of the main traditional argument is matched below to concrete formal declarations. The elementary normalization in Corollary 1.4 and the scope observations are derived here; the earlier geometric endpoint is already separately formalized. Proposition 9.1 is a traditional scope result, not claimed to be a Lean theorem in these frozen modules.

The available formal evidence has two levels. The original manuscript’s delivered harness and source correspondence are retained as historical records. The current repository additionally provides the full frozen proof release at `formalizations/continuum-remainder-avoidance`, with its public strict replay harness and explicit empty-environment kernel checker. The reproduction guide for this integrated manuscript identifies the exact targets, source hashes, toolchain and fresh replay records. A successful source-integrity check alone is not a proof-assistant replay.

A cache-miss exposed a dependency-parser defect in the release harness: Lean’s `importal1Module` syntax was parsed as importing a module named `all`. The pinned adapter `verification/finalization/replay_continuum.py` uses a separately SHA-pinned, comment/string-aware scanner from the simplex release, walks imports of cached modules to complete partial dependency caches, and supplies the pinned Mathlib Lake elaboration options `autoImplicit=false` and `maxSynthPendingDepth=3` when compiling missing Mathlib dependencies. It does not alter any mathematical source, release hash, target, axiom check, fresh-owned-module compilation rule, or kernel replay check. The historical package is left intact. The adapter is required when reproducing the proof from dependencies without a complete cache.

All fresh replay counts and outcomes belong to the corresponding machine-readable records; earlier reported kernel counts are not silently relabeled as new replays. No external human referee or professional mathematician is represented as having reviewed this manuscript.

The fixed environment is Lean 4.34.1 and mathlib commit `d13f23b723b8a846827a245b89c10fc7d3f11612`; all transitive revisions are retained in the supplied lockfile. The recorded axiom closures use only `propext`, `Classical.choice`, and `Quot.sound`. The paper’s traditional proof is independently readable and does not appeal to a proof-assistant theorem as an unproved mathematical premise.

Table 1: Traditional proof steps and exact frozen Lean declarations.

Paper step	Lean declarations and source modules
1.1. Logarithmic bins, per-configuration bounded gaps and exact endpoint specification	<code>ContinuumRemainder.OccupiedBin</code> <code>ContinuumRemainder.LogSyndetic</code> <code>ContinuumRemainder.ContinuumPowerTarget</code> <code>ContinuumRemainder.RobustCompactBlockerSpec</code> Modules: <code>ContinuumRemainder.Specification</code>
2.2. Separated sampling and strict output-window counts	<code>ContinuumRemainder.LogSyndetic.exists_sampledLogConfiguration</code> <code>ContinuumRemainder.open_window_count_bounds</code> <code>ContinuumRemainder.SampledLogConfiguration.activeLabels_count_bounds</code> Modules: <code>ContinuumRemainder.Sampling</code> ; <code>ContinuumRemainder.Counting</code>
3.1. Local span, finest endpoints and affine total span	<code>ContinuumGeometric.RoutingTemplate.blockSpan_le_twice_length</code> <code>ContinuumGeometric.RoutingTemplate.span_affine</code> <code>ContinuumGeometric.RoutingTemplate.edgeEnd_within_total</code> Modules: <code>ContinuumGeometric.RoutingTemplate</code> ; <code>ContinuumGeometric.RoutingTreeBounds</code>
3.2. Stable earlier addresses and distinct own-grid keys	<code>ContinuumRemainder.log_stable_earlier_address_agreement</code> <code>ContinuumRemainder.log_gridAddress_ne_center</code> <code>ContinuumRemainder.log_gridAddress_ne_of_output_gap</code> <code>ContinuumGeometric.unitDensity_routingUnstableCenters_le</code> Modules: <code>ContinuumRemainder.LogGeometry</code> ; <code>ContinuumGeometric.RoutingStableMeasure</code>
3. Actual finite global grid and Bernoulli terminal expectation	<code>ContinuumGeometric.actual_routedSet_finite_grid</code> <code>ContinuumGeometric.actual_routedSet_expected_density</code> Modules: <code>ContinuumGeometric.RoutingGlobalGrid</code> ; <code>ContinuumGeometric.FiniteRoutingProbability</code>
4. Full center exposure and exact no-default probability	<code>ContinuumGeometric.actual_center_atom_probability</code> <code>ContinuumGeometric.actual_center_atoms_sum_one</code> <code>ContinuumGeometric.actual_routeHasNoDefault_probability</code> Modules: <code>ContinuumGeometric.FiniteRoutingProbability</code> ; <code>ContinuumGeometric.NoDefaultProbability</code>
4.1. Joint atom miss identity without division or path independence	<code>ContinuumGeometric.joint_center_atom_all_miss</code> <code>ContinuumRemainder.log_fixed_parameter_joint_miss</code> Modules: <code>ContinuumGeometric.FiniteRoutingProbability</code> ; <code>ContinuumRemainder.LogRoutingProbability</code>

Paper step	Lean declarations and source modules
5.1. Boundary-complete planar affine sign arrangement	<code>ContinuumGeometric.realizedLinePatterns_card_le</code> <code>ContinuumGeometric.realizedPlanePatterns_card_le</code> <code>ContinuumGeometric.arrangementRepresentativeBound</code> Modules: <code>ContinuumGeometric.LineSignBound</code> ; <code>ContinuumGeometric.Planar</code>
5. Candidate count, activation signs and exact grid crossing signs	<code>ContinuumRemainder.SampledLogConfiguration.potentialPairs_card_le</code> <code>ContinuumRemainder.log_activation_cut_signs</code> <code>ContinuumRemainder.log_grid_boundary_sign</code> <code>ContinuumRemainder.log_grid_key_eq_of_log_signs</code> Modules: <code>ContinuumRemainder.Counting</code> ; <code>ContinuumRemainder.LogSignatures</code>
5.2. Representatives uniform in all outcomes and the continuum union bound	<code>ContinuumRemainder.log_local_grid_representatives_uniform</code> <code>ContinuumRemainder.log_representatives_entropy_bound</code> <code>ContinuumRemainder.log_continuum_joint_miss_bound</code> <code>ContinuumRemainder.sample_vertex_continuum_joint_miss_bound</code> Modules: <code>ContinuumRemainder.LogSignatures</code> ; <code>ContinuumRemainder.LogRoutingProbability</code> ; <code>ContinuumRemainder.SampleLocalProbability</code>
6.1. Logarithmic span, vanishing actual buffer cost and simultaneous late schedule	<code>ContinuumRemainder.scheduledSpan_isBig0_log</code> <code>ContinuumRemainder.sublinear_buffer_cost_tendsto_zero</code> <code>ContinuumRemainder.exists_late_robust_schedule</code> <code>ContinuumRemainder.exists_sample_routing_schedule_late</code> Modules: <code>ContinuumRemainder.ErrorSchedule</code> ; <code>ContinuumRemainder.SampleSchedule</code>
7. Active error domination and strict open double buffers	<code>ContinuumRemainder.active_power_error_lt_radius</code> <code>ContinuumRemainder.active_error_in_outer_buffer</code> <code>ContinuumGeometric.unitDensity_periodicGridSet_double_buffer_le</code> Modules: <code>ContinuumRemainder.ErrorDomination</code> ; <code>ContinuumGeometric.PeriodicRepair</code>
7.1. Closed missed centers, compact projection and stable probability	<code>ContinuumRemainder.isClosed_sampledMissedCenters</code> <code>ContinuumRemainder.sampled_stable_missed_center_probability_le</code> <code>ContinuumRemainder.sampled_stable_bad_density_le</code> Modules: <code>ContinuumRemainder.RobustRepair</code> ; <code>ContinuumRemainder.SampleStableProbability</code>
7.2. Open cover, every-center convergence and outcome assembly	<code>ContinuumGeometric.closed_onePeriodic_open_cover</code> <code>ContinuumRemainder.perturbed_power_tendsto</code> <code>ContinuumRemainder.sampled_robust_repair_all_centers</code> <code>ContinuumRemainder.actual_sampled_routing_blocker_of_stable_miss</code> Modules: <code>ContinuumGeometric.PeriodicRepair</code> ; <code>ContinuumRemainder.RobustRepair</code> ; <code>ContinuumRemainder.RobustAssembly</code>
2.1. Unconditional uniform robust blocker	<code>ContinuumRemainder.robustCompactBlockerSpec_proved</code> Modules: <code>ContinuumRemainder.FinalProof</code>

Paper step	Lean declarations and source modules
8. Countable exhaustion, coefficient normalization and reflected blockers	<code>ContinuumRemainder.robustCompactBlockerSpec_open_exhaustion</code> Modules: <code>ContinuumRemainder.Exhaustion</code>
1.2. Infinitely many distinct values, strict unit measure and nowhere density	<code>ContinuumRemainder.powerRemainder_two_sided</code> <code>ContinuumRemainder.infinite_tailValuesOutside_of_hits</code> <code>ContinuumRemainder.volume_onePeriodic_inter_Icc</code> <code>ContinuumRemainder.continuum_power_target</code> Modules: <code>ContinuumRemainder.DistinctMisses</code> ; <code>ContinuumRemainder.TopologyConclusion</code> ; <code>ContinuumRemainder.FinalProof</code>
1.3. Compact restriction with unchanged avoidance quantifiers	<code>ContinuumRemainder.compact_power_avoidance</code> Modules: <code>ContinuumRemainder.FinalProof</code>
1.4. Earlier independently proved all-ratio endpoint; current normalization is a traditional corollary	<code>ContinuumGeometric.MainTarget</code> <code>ContinuumGeometric.geometric_main_target</code> Modules: <code>ContinuumGeometric.Target</code> ; <code>ContinuumGeometric.MainProof</code>
A. Exact partial-tail/total-extension correspondence	<code>ContinuumRemainder.powerRemainderOn_totalTailExtension</code> <code>ContinuumRemainder.tailValuesOutside_totalTailExtension</code> Modules: <code>ContinuumRemainder.DistinctMisses</code>

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